High-Resolution Mapping of Dry and Wet Snow Using Multisource Remote Sensing Data in Mountainous Regions

Yonghui Zhang, Xiaodong Huang, Ying Ma, and Yuxin Li

***Abstract*—Accurate identification of dry and wet snow extents is crucial for runoff modeling and water resource management. This study, carried out in Qilian County from January 26 to May 2, 2020, introduces a novel approach that integrates optical and synthetic aperture radar (SAR) data to map snow cover conditions. Specifically, snow extent was derived from fused Moderate Resolution Imaging Spectroradiometer (MODIS) and Landsat-8 OLI data using the enhanced spatial and temporal adaptive reflectance fusion model (ESTARFM), which generated seven-day mean Landsat-8 simulated images with a spatial resolution of 30 m. The snow extent was classified using the SNOMAP algorithm. The wet snow extent was identified using a multitemporal, multipolarization change detection algorithm applied to Sentinel-1 SAR data, with local incidence angle (LIA) correction. Corrections for dry and wet snow extents were made by estimating the mean elevation of wet snow across different slope orientations. Ultimately, we produced maps of dry and wet snow extents at a spatial resolution of 30 m for every seven-day period. The results demonstrate that the fusion of MODIS and Landsat-8 images performs effectively, with R^2 values between the corresponding bands of simulated and real images exceeding 0.73. The extent of dry and wet snow in the study area exhibits significant variability with temperature and undergoes multiple melting processes. The method proposed in this study enables continuous and detailed monitoring of large-scale snowmelt processes, providing essential scientific support for extensive studies on energy, climate, and water cycles.**

***Index Terms*—Data fusion, dry and wet snow, Landsat, mapping, mountainous, Sentinel-1.**

I. Introduction

SNOW is an important indicator of global climate change and is one of the most active components of the cryosphere. Due to its highly reflective, widely distributed, and seasonally variable nature, snow reflects most of the solar radiation and consumes energy as it melts seasonally, thereby exerting a cooling effect on the Earth and affecting the exchange of energy and water vapor between the surface and the atmosphere [1], [2], [3]. In addition, snowmelt is a valuable freshwater resource, providing water for about 1/6 of the global population [4], [5], especially in arid and semi-arid regions where most freshwater resources come from seasonal snowmelt [6], [7]. However, with global warming, snow cover in the northern hemisphere has been shrinking and becoming less stable [8], [9], [10], which will lead to a series of ecological and environmental problems, including changes in snowmelt runoff [11] and frequent flooding [12], [13]. Accurate flood forecasting and water resource management in snow-dominated basins remain subject to several persistent challenges. In flood prediction, misclassification between dry and wet snow can result in underestimated peak flows or delayed flood warnings [14], [15]. In water resource management, the lack of timely and high-resolution snow condition data hampers the accurate estimation of seasonal water availability. Overestimation of dry snow may falsely suggest ample water reserves, while underestimation of early melt can disrupt reservoir operations and agricultural irrigation scheduling [16], [17]. Therefore, accurate information on dry and wet snow is crucial for runoff modeling, flood forecasting, and water resource management and utilization [18], [19], [20]. In snow-dominated mountainous catchments, changes in snow conditions directly influence flooding and the mechanisms of seasonal and interannual runoff variability. Consequently, changes in the extent of dry and wet snow significantly influence the accuracy of hydrological modeling [21], [22], [23]. Moreover, the extent of wet snow plays a vital role in the definition of seasonal parameters such as melt rate or albedo. It can also be used for calibration and validation of snowmelt processes to improve the accuracy of hydrological modeling results [24], [25], [26].

Information on dry and wet snow can be obtained through field surveys and snow data recorded by meteorological stations. However, in mountainous areas with complex terrain, data collection is challenging due to the sparse distribution of stations [27]. Remote sensing data provide critical support for obtaining this information in such regions [28], [29], [30]. Optical remote sensing satellites such as Advanced Very High-Resolution Radiometer (AVHRR), Moderate Resolution Imaging Spectroradiometer (MODIS), Sentinel-2, and Landsat are highly sensitive to the high reflectivity of snow. Typically, snow cover extent and its changes are detected using the ratio, difference, or the normalized difference snow index (NDSI) derived from the bands of these sensors [31], [32], [33]. Among these sensors, AVHRR and MODIS offer continuous

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snow cover information with high temporal resolution, but their low spatial resolution (500–1000 m) limits their ability to monitor snow cover in areas with complex local topography, resulting in generally low snow detection accuracy [33], [34]. In contrast, Sentinel-2 and Landsat satellites provide higher spatial resolution (10–20 m), but their longer revisit periods make it difficult to monitor rapid changes in snow cover [35]. Spatiotemporal fusion methods can combine the advantages of both types of data to generate remote sensing images with high spatiotemporal resolution [36], [37]. Traditional data fusion methods, however, cannot simultaneously improve both spatial and temporal resolution [38], [39], [40]. To address this limitation, a spatial and temporal adaptive reflectance fusion model (STARFM) was developed. This model fuses Landsat and MODIS data to predict surface reflectance on any given day under cloud-free conditions [41]. Nevertheless, the accuracy of this method is reduced in regions with complex terrain. The enhanced STARFM (ESTARFM) was later proposed to significantly improve fusion accuracy [42]. This fusion algorithm is now widely applied in studies related to vegetation, surface temperature, and lake monitoring [43], [44], [45], [46].

The signals received by both active and passive microwave remote sensing sensors are influenced by the physical properties of snow [47]. Compared to passive microwave remote sensing, active microwave remote sensing offers higher spatial resolution and all-weather, all-time observation capabilities, making it more suitable for monitoring changes in wet snow cover [48]. Studies have demonstrated that the water content in wet snow inhibits the penetration of electromagnetic waves, resulting in a significant decrease in the backscattering coefficient [49], [50], [51].

A radar snow fraction (RSF) index based on fully polarized synthetic aperture radar (SAR) data has been proposed. This index enables efficient snow cover identification and mapping without the need for multitemporal data by combining entropy and anisotropy parameters [52]. In addition, many studies distinguish dry and wet snow by comparing the backscatter intensity of SAR snowmelt images with that of reference images from dry snow or snow-free periods [33], [53], [54]. An SAR multitemporal change detection algorithm has been proposed for wet snow identification [55]. Building on this approach, researchers have proposed an SAR multitemporal multipolarization change detection algorithm, which improves the accuracy of wet snow cover extraction in mountainous areas by integrating the weights of VV and VH polarizations and the local incidence angle (LIA) [56].

Currently, several studies have combined optical and active microwave remote sensing data to identify dry and wet snow extent. A study employed support vector machine (SVM) and multilayer perceptron (MLP) regression models integrating Sentinel-1 and MODIS data to develop a high-resolution snow cover area (SCA) mapping framework, which significantly enhances the accuracy of snow classification [57]. Another study monitoring spring snowmelt processes in northern Finland using ERS-2 SAR and AVHRR data demonstrated that combined SAR and optical remote sensing data can accurately monitor changes in snowmelt in areas without ground cover and in vegetated areas [58]. In addition, several studies have obtained dry and wet snow extent in the study area using Radarsat-2 SAR or Sentinel-1 SAR to extract wet snow extent and MODIS to extract total snow extent [27], [38]. However, these studies encountered different degrees of error in distinguishing dry and wet snow extent due to the limited resolution of optical satellite data. To address this limitation, some researchers have turned to high-resolution optical satellite data for snow cover extraction. A study on the distribution of wet and dry snow in the Solang valley of the Himalayas was conducted to extract wet snow using the SAR change detection algorithm based on Sentinel-1 SAR data and snow extent based on Sentinel-2 multispectral data by calculating NDSI [59]. In recent years, some studies have utilized the ESTARFM fusion algorithm to fuse Sentinel-2 and MODIS data to identify the snow cover and utilized Sentinel-1 SAR to obtain wet snow extent, enabling rapid monitoring of snowmelt changes [53], [60]. Nevertheless, the applicability of these methods in large-scale complex terrain areas remains uncertain.

In view of the above, based on the Google Earth Engine (GEE) platform, this study proposes a novel fusion strategy using the ESTARFM method to fuse MODIS and Landsat-8 OLI multiday mean images to obtain seven-day mean Landsat-8 simulated images with 30-m spatial resolution, and to obtain snow extent based on the SNOMAP algorithm. In addition, LIA is combined with Sentinel-1 SAR data through a multitemporal, multipolarization SAR change detection algorithm to obtain wet snow extent at the same spatial resolution. Furthermore, by estimating the average elevation of wet snow across different slopes, the boundaries between dry and wet snow are corrected, ultimately producing a 30-m spatial resolution map of dry and wet snow extent for Qilian County from January 26 to May 2, 2020. This method can also be extended to any region worldwide with Sentinel-1 SAR data, enabling continuous and detailed monitoring of the large-scale snow ablation process in different regions of the world and providing scientific support for the study of large-scale energy, climate, and water cycles.

II. Study Area and Data

A. Study Area

The study area is Qilian County, located in the northwest of the Tibetan Plateau (37°N, 98°E). The region is located in the hinterland of Qilian Mountains, with an average annual temperature of −1.8 °C and annual precipitation ranging from approximately 272.7 to 650.7 mm. Qilian County spans approximately 185 km from north to south and 280 km from east to west, with a total area of 13919.98 km^2[61]. The elevation ranges from 2274 to 5274 m, with an average elevation of 3169 m. The major mountain ranges from north to south include the north of Tuole Mountain, Tuole Mountain, the north of Zoulang Mountain, and the western section of the Lenglongling(Fig. 1).

There are three major rivers in Qilian County: the Datong River of the Yellow River system, the Hei River, China's second largest inland river, and the Beida River. The Datong River flows through Qilian County, while the Hei River originates from the Bayi Glacier, and the Tuole River comes from the Qiyi Glacier. These rivers are created by the melting

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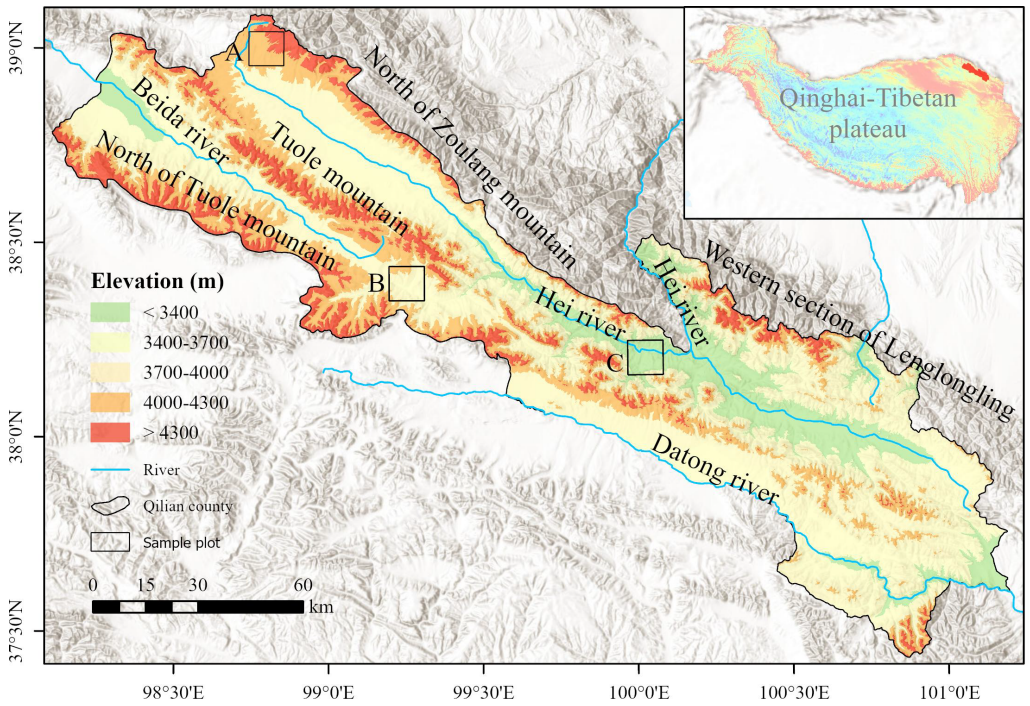

Fig. 1. Overview of the study area. Three sample Plots A–C were used for accuracy validation of the ESTARFM fusion model.

glaciers and snow [62], [63]. Statistics indicate that the annual runoff of rivers in Qilian County is 23.22×10^9 m^3, with a theoretical hydropower resource potential of 56.77×10^4 kw [64]. Overall, Qilian County possesses abundant freshwater and hydropower resources, so the study of dry and wet snow extent in the region is crucial for understanding energy, environmental, and climatic effects, as well as for hydrological modeling, water resource management, and hydropower development in the area.

B. Data and Preprocessing

1) Optical Remote Sensing: In this study, Sentinel-2 MSI and Landsat-8 OLI data were used to validate the accuracy of the proposed model in identifying SCA. Landsat-8 OLI and MOD09GA data were selected for data fusion, utilizing three spectral bands: green, near-infrared (NIR), and shortwave infrared (SWIR). Sentinel-2 has a spatial resolution of 10 m and a temporal resolution of five days, Landsat-8 has a spatial resolution of 30 m and a temporal resolution of 16 days, while MOD09GA has a spatial resolution of 500 m and a temporal resolution of one day. Due to frequent cloud and snow cover in mountainous regions, acquiring temporally aligned and cloud-free imagery from multiple sensors remains challenging. To address this, 30- and 45-day multiday mean reflectance composites were generated for both Landsat-8 and MOD09GA, serving as inputs to the ESTARFM model. These composites enable the production of seven-day mean simulated Landsat-8 images with enhanced spatial and temporal consistency.

The choice of 30- and 45-day time windows is based on actual data availability and the temporal dynamics of snow cover. Extended time windows increase the number of valid observations, which improves with model training and reduces the impact of cloud contamination. At the regional scale, snow accumulation and melt typically exhibit gradual seasonal changes [65]. Therefore, a window length of 30 and 45 days is sufficient to capture the main temporal trends while minimizing the influence of short-term events such as transient snowfall or melt episodes. The ESTARFM model assumes that surface reflectance changes linearly between observation dates [42]. Excessively long-time spans may violate this assumption, especially in areas undergoing rapid surface changes. Thus, the 30- and 45-day windows offer a practical balance between temporal smoothing and the model's fundamental assumptions.

This study utilizes the GEE, a cloud-based integrated development environment (IDE), to preprocess and download Landsat-8 OLI and MOD09GA data. GEE is a cloud analysis platform for Earth observation data [66], providing global-scale data products such as Landsat-8 OLI and MOD09GA. First, the study acquired Landsat-8 OLI and MODIS data, primarily performing cloud removal on MOD09GA, while the processing of Landsat-8 OLI data was more complex. Compared to Landsat-7 ETM+, Landsat-8 OLI offers several advantages, including narrower spectral bands, improved calibration and signal-to-noise ratio, higher 12-bit radiometric resolution, and more accurate geometry [67], [68], [69], [70]. Therefore, this study employed Landsat-8 OLI data for the simulation research. Initially, ESTARFM was used to fuse MODIS and Landsat-7 ETM+. However, using sensors with different spectral bands in the ESTARFM model may introduce nonlinear relationships, so the three bands of the declouded Landsat-8 OLI are corrected to be consistent with the Landsat-7 ETM+ [42]. We transformed the band values of Landsat-8 OLI by using the statistical function between Landsat 7 ETM+ and Landsat-8 OLI band values. This transformation function, developed using ordinary least-squares

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(OLS) regression, provides a highly reliable fit [68]. This conversion ensures better compatibility of Landsat-8 OLI data with the ESTARFM model. Furthermore, although using long-term multiday averages reduces gaps in the Landsat-8 images for the study area, some regions still exhibit gaps due to cloud removal. Thus, this study uses MOD09GA mean images from the same time period to fill the gaps in the Landsat-8 mean images.

2) SAR Data: This study selected the ascending and descending orbit data (VV and VH polarization) from the interferometric wide (IW) swath mode of Sentinel-1A GRD products, with a revisit period of 12 days. The data were obtained from the GEE dataset, which provides preprocessed Sentinel-1 SAR imagery. These images were processed using SNAP software to apply precise orbit file corrections, remove border noise, eliminate thermal noise, perform radiometric calibration, and apply terrain Doppler correction. Finally, the backscatter coefficient σ^0 was converted to dB (i.e., by $10 \times \log_{10}\sigma^0$ is converted to dB). To match the spatial resolution of the simulated Landsat-8 OLI data, the SAR data were resampled to a 30-m spatial resolution.

Due to the limited revisit frequency of Sentinel-1, capturing the full extent of wet snow cover within a seven-day interval may be challenging. Therefore, descending orbit data were used as a supplement. The acquisition time of the images also affects the detection of wet snow. Since Sentinel-1 images are acquired at 11:19 (ascending orbit) and 23:27 (descending orbit), the descending orbit acquisition occurs at night when temperatures are relatively low, which may result in Sentinel-1 missing some areas of wet snow. The method developed to correct Sentinel-1's underestimation of wet snow extent is described in detail in Section III-E.

3) Auxiliary Data: This study also acquired the digital elevation model (DEM) and land cover type data for the study area to analyze the variability of dry and wet snow extent, calculate the LIA of SAR data, and implement corrections for dry and wet snow extent. The DEM used in this study was released by European Space Agency (ESA) through the Copernicus program (https://scihub.copernicus.eu/), with a spatial resolution of 30 m. The land cover type data were sourced from the ESA World Cover product (https://esa-worldcover.org), provided by European Copernicus program, with a resolution of 10 m. This dataset represents global land cover types for 2021, derived from Sentinel-1 and Sentinel-2 data. All the above data were resampled to a spatial resolution of 30 m (see Table I).

III. Methods

In this study, preprocessed optical remote sensing data were fused to obtain the seven-day mean Landsat-8 simulated image with 30-m spatial resolution using the ESTARFM algorithm. The snow extent was obtained based on the SNOMAP algorithm, and the wet snow extent was identified based on Sentinel-1 SAR data and the corresponding LIA. We also corrected the dry and wet snow extent by estimating the average elevation of wet snow across different slope orientations. Finally, a 30-m resolution map of dry and wet snow extent in Qilian County was generated for every seven-day period (Fig. 2).

A. ESTARFM Fusion Algorithm

ESTARFM is an algorithm developed to fuse high spatial resolution and high temporal resolution remote sensing data using multisource satellite data [38]. This section focuses on fusing multiday mean data from Landsat-8 OLI and MOD09GA using ESTARFM. Without considering atmospheric correction and registration errors, the ESTARFM method establishes a mathematical relationship between the 30-day mean images of MOD09GA and Landsat-8 and the seven-day mean images of MOD09GA to obtain the seven-day mean simulated Landsat-8 image (Fig. 3).

The implementation of the ESTARFM algorithm consists of four major steps. First, MODIS and Landsat-8 need to undergo preprocessing. MODIS images are cloud-masked and resampled to 30-m spatial resolution. Landsat-8 images are processed through cloud removal, OLS regression, and gap filling. Second, a pair of fine-resolution Landsat images is used to search for spectrally similar pixels within a local window. Third, the weights of all similar pixels (W_i) are calculated, and the conversion coefficients V_i are determined by linear regression. Finally, W_i and V_i are applied to the coarse-resolution MODIS data to predict fine-resolution surface reflectance at the target date (Fig. 4) [42].

During the prediction process, the sliding window method is used to reduce the effect of pixel boundaries from low-resolution remote sensing data. When calculating the central pixels using the sliding window, spatial distance, spectral distance, and temporal distance are employed as weights. The prediction algorithm for the seven-day mean image and 30-day mean image using ESTARFM can be expressed by the following equation:

$$F\left(x_{\partial/2}, y_{\partial/2}, t_7, B\right) = F\left(x_{\partial/2}, y_{\partial/2}, t_{30}, B\right) + \sum_{i=1}^{n} W_i \times V_i \times \left(C\left(x_i y_i t_7, B\right) - C\left(x_i y_i t_{30}, B\right)\right) \quad (1)$$

where $F(x_{\partial/2}, y_{\partial/2}, t_7, B)$represents the predicted pixel value of the seven-day high-resolution mean image, while $F(x_{\partial/2}, y_{\partial/2}, t_{30}, B)$ represents the pixel value of the 30-day high-resolution mean image. $(x_{\partial/2}, y_{\partial/2})$ is the central pixel of the window. n is the number of similar pixels for the central predicted pixel. $C(x_i, y_i, t_7, B)$ and $C(x_i, y_i, t_{30}, B)$ are the pixel values of the seven-day coarse-resolution mean image and the 30-day coarse-resolution mean image, respectively. W_i is the conversion coefficient for the ith similar pixel. W_i is the weight for the ith similar pixel. B is the band chosen for fusion. The prediction algorithms of ESTARFM for the 45-day mean image and the seven-day mean image are similar to those for the seven-day mean image and the 30-day mean image

$$F\left(x_{\partial/2}, y_{\partial/2}, t_7, B\right) = F\left(x_{\partial/2}, y_{\partial/2}, t_{45}, B\right) + \sum_{i=1}^{n} W_i \times V_I \times \left(C\left(x_i y_i t_7, B\right) - C\left(x_i y_i t_{45}, B\right)\right). \quad (2)$$

Finally, the seven-day mean image of Landsat-8 can be obtained by a weighted summation of the two fusion results. The weighted β_t is shown in (2), and the seven-day mean image of Landsat-8 is merged using β_t, expressed as in (3) and (4), shown at the bottom of the next page.

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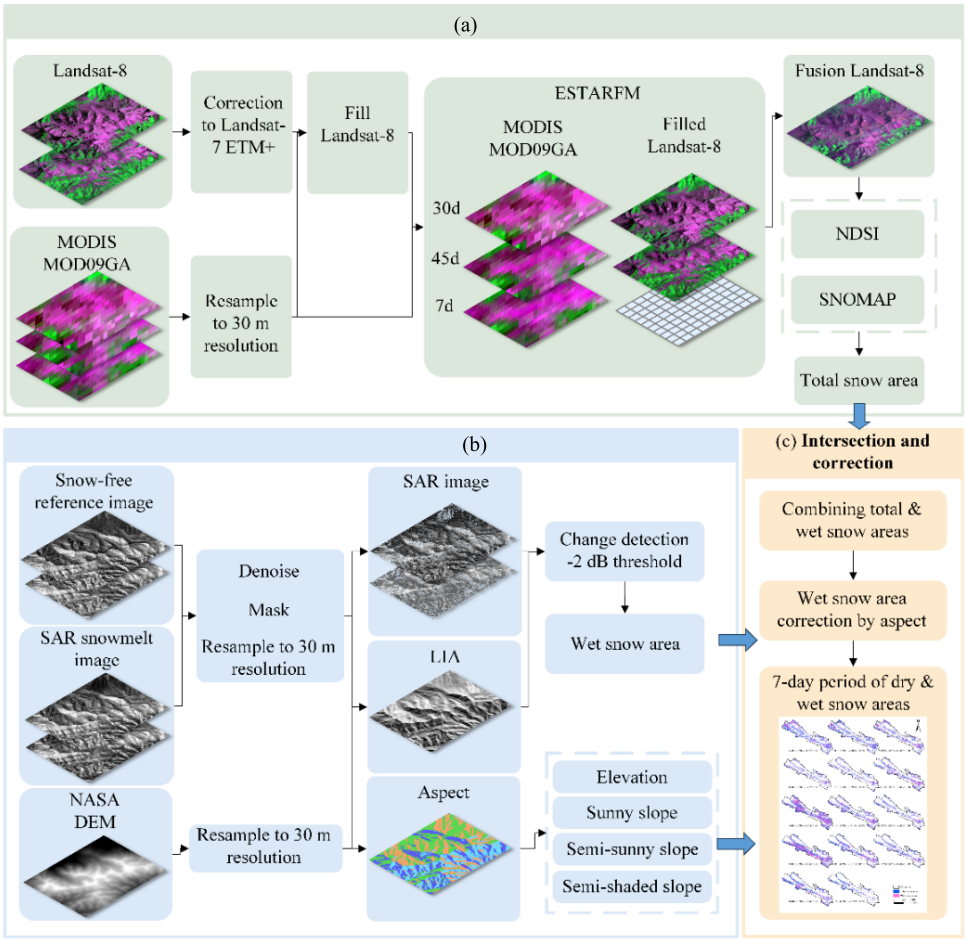

Fig. 2. Flowchart of spatiotemporal data fusion and dry and wet snow extent monitoring. (a) Total snow area detection. (b) Wet snow area detection. (c) Intersection and correction.

B. Snow Cover Mapping

In this study, the SNOMAP algorithm was used to extract SCA from Landsat-8, Sentinel-2, and simulated Landsat-8 imagery. The SNOMAP algorithm is a snow identification algorithm based on snow cover characteristics [33]. The algorithm adds threshold method discriminant functions for NIR and green wave bands to the NDSI, which can effectively reduce the probability of misclassification of snow extent in water bodies, dense vegetation, shadows, and low light conditions [36]. The calculation formula is as follows:

$$\begin{cases} \mathrm{NDSI} = \dfrac{\mathrm{Band}_{\mathrm{Green}} - \mathrm{Band}_{\mathrm{SWIR1}}}{\mathrm{Band}_{\mathrm{Green}} + \mathrm{Band}_{\mathrm{SWIR1}}} \geq 0.40 \\ \mathrm{Band}_{\mathrm{NIR}} \geq 0.11 \\ \mathrm{Band}_{\mathrm{Green}} \geq 0.10. \end{cases} \tag{5}$$

C. Comparison

This study employed three methods to validate the simulated Landsat-8 data generated using the ESTARFM model. First, the spectral accuracy of the simulated images was assessed by comparing the simulated Landsat-8 data with actual Landsat-8 observations across three spectral bands. The evaluation metrics included standard deviation(std), information entropy (ie), correlation coefficient(R), and significance level (p-value). Second, SCA derived from actual Landsat-8 and Sentinel-2 imagery was used as reference data. Based on the confusion matrix method, the accuracy of snow cover identification using the simulated Landsat-8 data was evaluated. Finally, simulated Landsat-8 data were also generated using the STARFM and FSDAF models, both of which rely on imagery from two time points [41], [71]. SCA was extracted from these simulations and evaluated using the confusion

$$\beta_t = \frac{1 \Big/ \left| \sum_{j=1}^{\partial} \sum_{i=1}^{\partial} C\left(x_i, y_j, t_\beta, B\right) - \sum_{j=1}^{\partial} \sum_{i=1}^{\partial} C\left(x_i, y_j, t_7, B\right) \right|}{\sum_{t=c,} \left(1 \Big/ \left| \sum_{j=1}^{\partial} \sum_{i=1}^{\partial} C\left(x_i, y_j, t_\beta, B\right) - \sum_{j=1}^{\partial} \sum_{i=1}^{\partial} C\left(x_i, y_j, t_7, B\right) \right| \right)}, \quad (t = c, e) \tag{3}$$

$$F_7\left(x_{\frac{\partial}{2}}, y_{\frac{\partial}{2}}, t_7, B\right) = \beta_c \times F_{30}\left(x_{\frac{\partial}{2}}, y_{\frac{\partial}{2}}, t_7, B\right) + \beta_e \times F_{45}\left(x_{\frac{\partial}{2}}, y_{\frac{\partial}{2}}, t_7, B\right) \tag{4}$$

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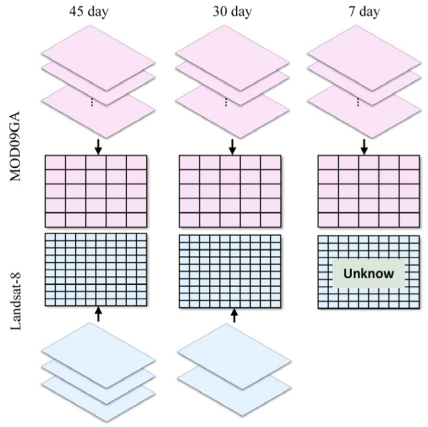

Fig. 3. Diagram of ESTARFM fusion algorithm for seven-day mean simulated Landsat-8 image.

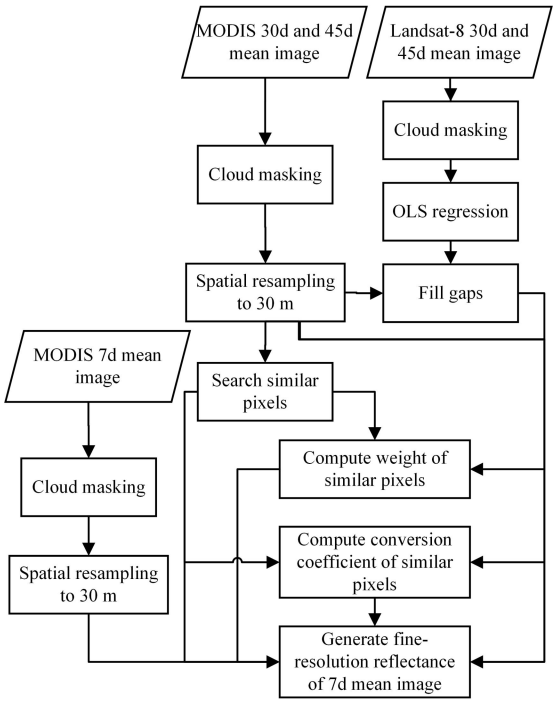

Fig. 4. Flowchart of the algorithm based on ESTARFM.

matrix method and then compared with the snow cover results derived from the ESTARFM. A confusion matrix was used for accuracy assessment in this study. Five evaluation metrics were applied: overall accuracy (OA), producer's accuracy (PA), user's accuracy (UA), Kappa coefficient, and Bias. The specific calculation formulas are as follows:

$$\mathrm{OA} = \frac{\mathrm{SS}+\mathrm{NN}}{\mathrm{SS}+\mathrm{NS}+\mathrm{SN}+\mathrm{NN}} \times 100 \tag{6}$$

$$\mathrm{PA} = \frac{\mathrm{SS}}{\mathrm{SS}+\mathrm{SN}} \times 100 \tag{7}$$

$$\mathrm{UA} = \frac{\mathrm{SS}}{\mathrm{SS}+\mathrm{NS}} \times 100 \tag{8}$$

$$\mathrm{Kappa} = \frac{(\mathrm{SS}+\mathrm{NS})(\mathrm{SS}+\mathrm{SN})+(\mathrm{NN}+\mathrm{NS})(\mathrm{NN}+\mathrm{SN})}{(\mathrm{SS}+\mathrm{SN}+\mathrm{NS}+\mathrm{NN})^2} \tag{9}$$

$$\mathrm{Bias} = \frac{\mathrm{SS}+\mathrm{NS}}{\mathrm{SS}+\mathrm{SN}} = \frac{\mathrm{PA}}{\mathrm{UA}}. \tag{10}$$

Accuracy (UA) is the ratio of correctly classified snow pixels to the total number of pixels identified as snow. The Kappa coefficient is used to measure the consistency between the reconstructed SCA and the actual SCA. Bias is a metric used to evaluate the overestimation or underestimation of the reconstructed SCA; a Bias > 1 indicates overestimation, while a Bias < 1 indicates underestimation. SS denotes the pixels predicted as snow and actually snow. NN denotes the pixels predicted as nonsnow and actually nonsnow. NS denotes the pixels predicted as snow but actually nonsnow. SN denotes the pixels predicted as nonsnow but actually snow.

D. LIA Acquisition

Extracting the wet snow extent requires information about the LIA, which is the angle between the incident vector and the surface normal vector [72], as shown in Fig. 4. Currently, GEE does not provide bands for obtaining LIA, so this study calculates the LIA based on a previous study by using GEE [73].

The formula for calculating LIA is based on the angle between the incident vector s and the surface normal n [74]. In addition, calculating LIA requires the slope α, azimuth angle β, and the sensor's observation azimuth angle γ for the study area. After derivation, (6) can be used to calculate LIA

$$\cos\theta_0 = \cos\theta \times \cos\alpha + \sin\alpha \times \sin\theta \times \cos(\gamma - \beta) \tag{11}$$

where θ_0 is the LIA, θ is the radar incidence angle, α is the slope, β is the azimuth angle, and γ is the sensor's observation azimuth angle. The radar incidence angle θ for each pixel is known. The slope $\alpha$and azimuth angle β were calculated using NASADEM [74] and then reprojected to match the projection of Sentinel-1 and resampled to a resolution of 30 m using nearest neighbor interpolation. The sensor's observation azimuth angle γ for each image is calculated separately, as it varies for each ascending track image. The observation azimuth angle γ is perpendicular to the satellite's flight direction relative to true north (i.e., the flight azimuth). The flight azimuth is derived by determining the corner points on the image, which are points on a vector parallel to the flight direction [74].

Equation (6) assumes that the incident vector s and the surface normal n have the same direction (see Fig. 5). To account for the original direction of the incident vector, we need to subtract 180° from the sensor's observation azimuth angle γ, yielding $\gamma' = \gamma - 180°$, to obtain the actual LIA [72], [73]. The formula can be expressed as

$$\cos\theta_0 = \cos\theta \times \cos\alpha - \sin\alpha \times \sin\theta \times \cos(\gamma' - \beta). \tag{12}$$

E. Wet Snow Detection and Correction

This study employed a multitemporal, multipolarization SAR change detection algorithm to extract wet snow extent. The algorithm detects wet snow by calculating the difference in backscatter coefficients between snowmelt-period images and reference images representing dry snow or snow-free conditions [75] (see Fig. 6). Typically, an increase in the liquid water content of snow leads to a reduction in the backscatter

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coefficient by approximately 1–3 dB [56], [76], [77]. In this study, a threshold of −2 dB was applied to identify wet snow. This threshold has been widely adopted in previous studies, particularly for snowmelt monitoring and wet snow detection in mountainous or cold regions [30], [56], [65], [76], [77], [78]. However, no image in the study area was found to be completely snow-free. Based on observations from optical imagery, three dates with minimal snow coverage were identified: August 18, August 30, and September 11 for ascending orbits; and August 23, September 4, and September 16 for descending orbits. These dates were selected as reference periods, and the mean backscatter coefficient calculated from them was used as the reference image.

In IW mode, Sentinel-1's incidence angle ranges from 29°to 46°. However, due to the complex terrain in the study area, the LIA range is expanded [30]. Previous studies have shown that at extreme LIA values (below 20° or above 70°), the difference in backscatter signals between wet snow and nonwet snow areas diminishes [30], [56], [78]. Therefore, this study masks areas with LIA (θ) less than 18° or greater than 78° [25]. In addition, denoising was applied to the VV and VH polarization backscatter coefficients by masking areas where the VV backscatter coefficient is less than −20 dB. The VH backscatter coefficient is greater than −30 dB and less than −24 dB [79]. The change detection images for VV and VH polarizations were obtained by calculating the difference in backscatter coefficients between the snowmelt and reference images, as shown in(8). Furthermore, since SAR cross-polarization compensates for reduced sensitivity in extracting wet snow in steep terrain [56], the VV and VH polarization backscatter coefficients were combined using a weighted approach, as shown in (9)

$$R = \sigma_{\mathrm{ws}} - \sigma_{\mathrm{ref}} \tag{13}$$

$$R_C = WR_{\mathrm{VH}} + (1 - W)\,R_{\mathrm{VV}} \tag{14}$$

where σ_{ws} and σ_{ref} are the backscatter coefficients of the snowmelt image and reference image, respectively; R is the backscatter coefficient of the change detection image; R_C is the backscatter coefficient from the weighted combination; and W is a weighting factor, which varies with the LIA [56], as shown in (10)

$$\begin{cases} \mathrm{IF}\,(\theta < \theta_1) \rightarrow W = 1.0 \\ \mathrm{IF}\,(\theta_1 \le \theta \le \theta_2) \rightarrow W = K\left[1 + \dfrac{(\theta_2 - \theta)}{(\theta_2 - \theta_1)}\right] \\ \mathrm{IF}\,(\theta > \theta_2) \rightarrow W = K \end{cases} \tag{15}$$

where K is a constant set at 0.5, θ_1 is 20°, and θ_2 is 45°. Based on the above formulas, the wet snow coverage can be obtained. Thanks to the computation of the LIA implemented in GEE, as described in Section III-C, the process of extracting wet snow coverage can also be carried out in GEE.

Research has shown that the wet snow information extracted by SAR contains certain uncertainties. Outside the snow-covered areas, Sentinel-1 may still detect wet snow, which may be due to the multitemporal, multipolarization SAR change detection algorithm or misclassification of surface water from snowmelt as wet snow [27]. In addition, trees and tree canopies can cause variations in backscatter coefficients, which may also be identified by the wet snow algorithm, particularly at lower thresholds [27]. In the study area, forests account for less than 0.9% of the region and have a minimal impact. Therefore, this study uses the snow coverage extracted from the simulated Landsat-8 images as the baseline, considering any wet snow outside the snow coverage area as misclassified pixels.

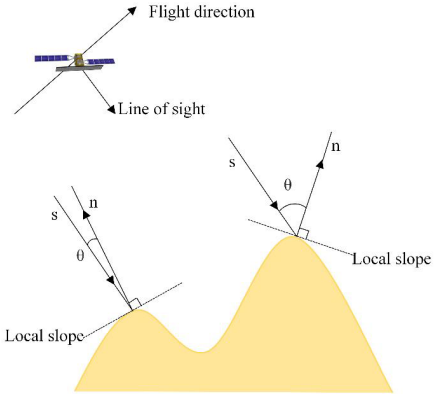

Fig. 5. Schematic of the LIA, where s is the incident vector; n is the vector perpendicular to the local slope, i.e., the surface normal; and θ is the local angle of incidence.

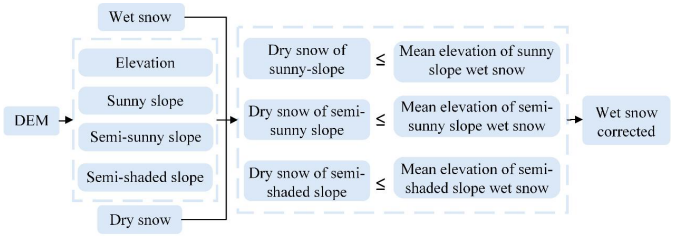

Fig. 6. Dry and wet snow extent correction method.

Since the temperature varies across slopes due to the solar radiation, the elevation of the dry and wet snow boundaries also differs. Thus, this study corrects the dry and wet snow extent derived from the combined Sentinel-1 and simulated Landsat-8 images using DEM data. The method for correcting the wet and dry snow extent with DEM involves first calculating the average elevation of wet snow for sunny slopes, semi-sunny slopes, and semi-shady slopes (with flat areas also classified as semi-sunny slopes). Next, dry snow pixels below the average elevation of wet snow are categorized based on whether they are below the average elevations for sunny, semi-sunny, or semi-shady slopes. Finally, all dry snow pixels below the average elevation of wet snow are corrected to wet snow pixels.

IV. Results

A. Evaluation of Data Fusion

This study analyzed the consistency between real and simulated Landsat-8 images from January 26 to February 1, 2020, when there was almost no cloud cover, in three typical areas: mountainous, plain, and watershed (shown as A–C in Fig. 1). The simulated images were generated using the ESTARFM fusion method. Consistency analysis was performed based on 10%of the total pixels in each image. As shown in Fig. 7, the scatter plots for the nine bands exhibit a concentrated distribution, with most points located on either side of the trend line.

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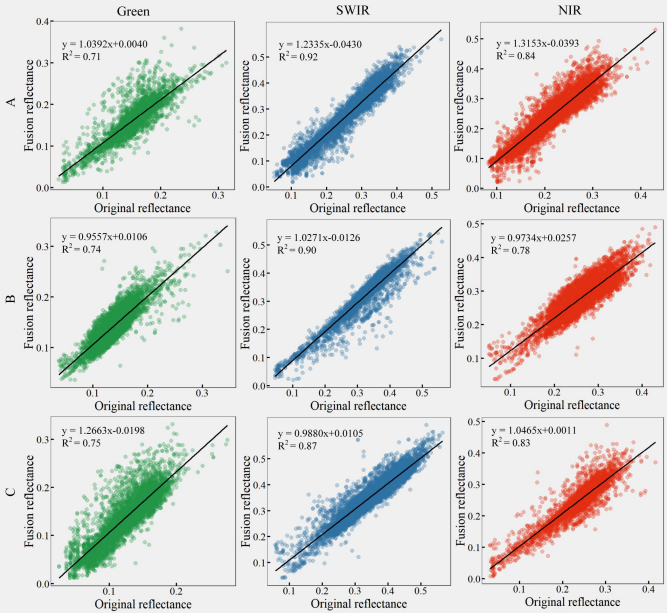

Fig. 7. Band scatter plots of green, SWIR1, and NIR for real and simulated images of sample Plots A–C.

TABLE I
CHARACTERISTICS OF DATA USED IN THIS STUDY

Dataset	Source	Spatial resolution	Temporal resolution	Usage
Sentinel-1 GRD	ESA	5m(azimuth)* 20m(range)	12 d	WSCA
Landsat-8 OLI	USGS	30m	16 d	ESTARFM Comparation
MOD09GA	NASA	500m	1d	ESTARFM
Sentinel-2	ESA	10-60m	5d	Comparation
ESA World Cover	ESA	10m	——	Correction
DEM	ESA	30m	——	Correction LIA

There is a very high consistency between the corresponding bands of real and simulated images, with R^2 values exceeding 0.71, indicating a strong correlation in reflectance between the simulated and real images. These results demonstrate that the ESTARFM algorithm yields high-quality simulated remote sensing images, which can be used for subsequent snow cover analysis.

In addition, we also qualitatively assess the image fusion effect by calculating the correlation coefficient (R), information entropy (ie), and standard deviation (std), and of the fused images compared with the real images in the three sample sites. As shown in Table II, the fused data from the three sample sites exhibit excellent fusion performance. The standard deviation and information entropy are very close to the real image with only minor differences. The correlation

TABLE II
STATISTICAL TABLE OF DATA FUSION RESULTS

Plots	Index	Real images	Fusion images
A	*std*	0.0870	0.0873
B		0.0825	0.1020
C		0.1037	0.1078
A	*ie*	13.82	13.82
B		13.78	13.76
C		13.72	13.79
A	*R*	0.96 (p<0.01)	
B		0.93 (p<0.01)	
C		0.97 (p<0.01)	

TABLE III
VALIDATION OF SCA DERIVED FROM ESTARFM

	OA%	PA%	UA%	Kappa	Bias
Landsat-8	89.12	88.75	87.3	0.801	1.017
Sentinel-2	86.95	86.42	85.1	0.765	1.014
Total	88.03	87.59	86.2	0.783	1.016

coefficients between the fused and real images are all above 0.92, indicating a strong correlation between the images. Therefore, the fusion using the ESTARFM algorithm is highly effective.

The SCA derived by the ESTARFM was validated against the SCA derived from both Landsat-8 and Sentinel-2 imagery (Table III). When validated with Landsat-8 data, OA reached 89.12%, while validation with Sentinel-2 data yielded an OA of 86.95%, indicating a high overall classification accuracy and robust performance. PA was 88.75% and 86.42%, suggesting that the missed classification of the snow area for the derived snow area was approximately 11%–14%, with relatively few missed snow pixels. UA was 87.30% and 85.10%, indicating that the error of misclassification was in the range of 13%–15%, which is within an acceptable range and reflects good consistency in snow cover detection. The Kappa was 0.801 and 0.765, both close to 0.8, demonstrating strong agreement between the reconstructed and actual snow distributions. For the Bias, the overall value was 1.016, indicating a slight overestimation of snow area compared to the actual snow cover, but the error was minimal. Overall, although the validation results based on Sentinel-2 were slightly lower than those based on Landsat-8, the difference was within 3%, demonstrating that the model maintains stable and reliable detection performance across different data sources.

By comparing the validation results of SCA derived using ESTARFM, FSDAF, and STARFM, it is evident that the SCA extracted from Landsat-8 imagery simulated by the ESTARFM model demonstrates overall higher accuracy than the other two methods, and shows good stability across different terrain regions (Table IV and Fig. 8). As shown, ESTARFM consistently achieves higher OA across various validation images,

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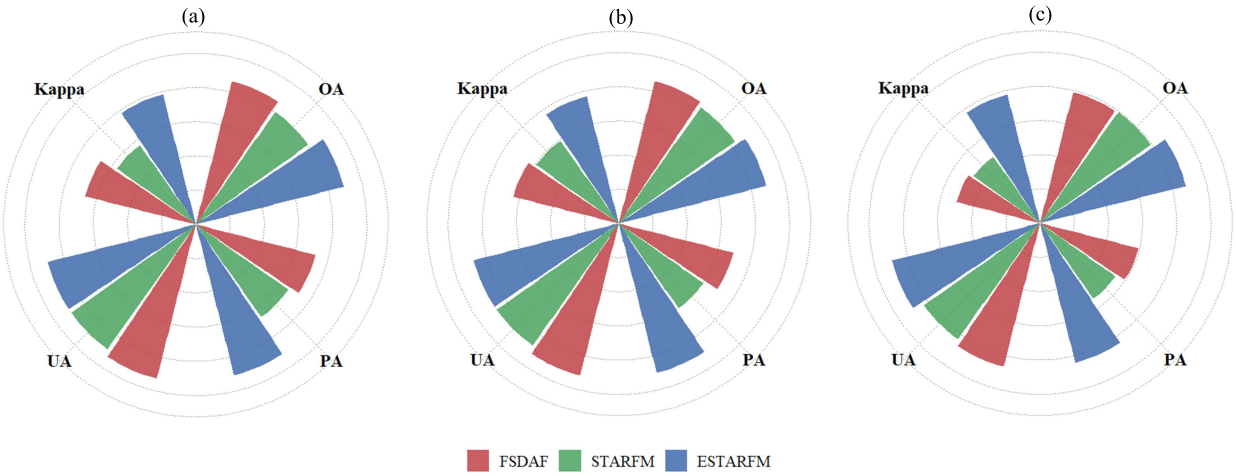

Fig. 8. Comparison of SCA derived by different models. (a)–(c) Sample plots A, B, and C, respectively.

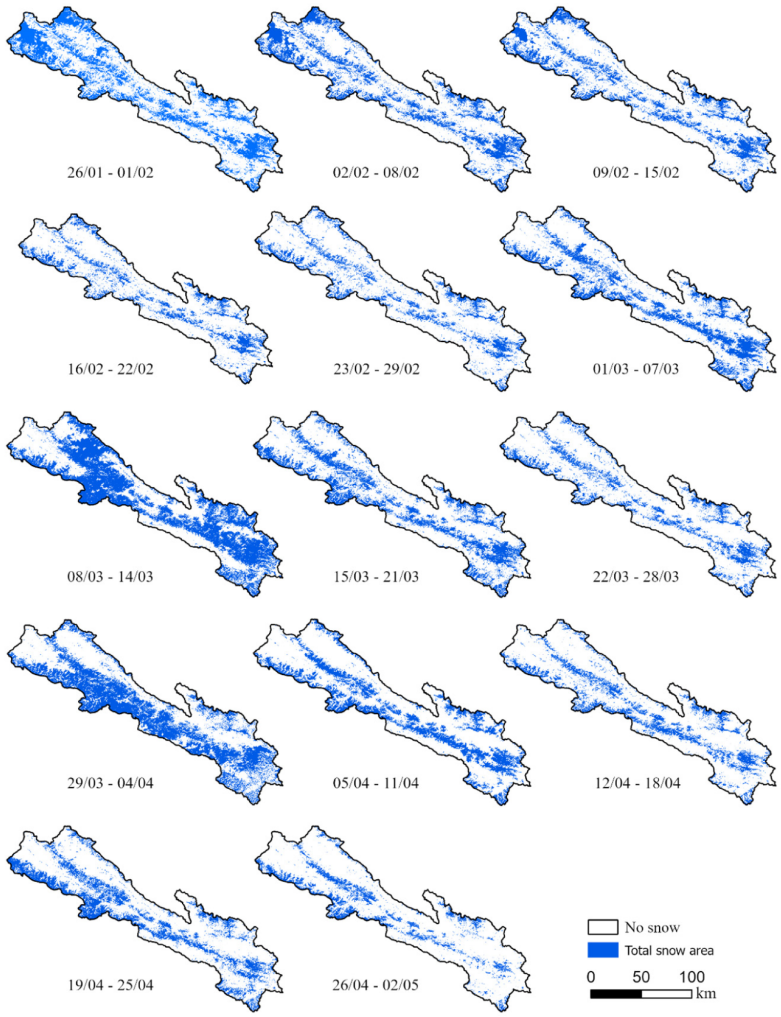

Fig. 9. Dry snow distribution in the study area from January 26 to May 2, 2020, based on simulated Landsat-8 data.

indicating better robustness. In contrast, OA values for SCA obtained from FSDAF and STARFM models are lower and exhibit significant fluctuations in Plot C (79.2% and 79.54%, respectively), reflecting reduced model stability. Moreover, the PA of the ESTARFM model is 18.78%–30.2% higher than that of the FSDAF and STARFM models, indicating

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TABLE IV
COMPARISON OF SCA DERIVED BY DIFFERENT MODELS

Plot	Model	OA%	PA%	UA%	Kappa	Bias
A	FSDAF	86.24	72.21	**92.89**	66.70	0.777
	STARFM	80.51	66.23	89.21	56.43	0.743
	ESTARFM	**89.33**	**90.99**	89.26	**78.24**	**1.01**
B	FSDAF	85.80	69.21	**91.79**	63.68	0.754
	STARFM	83.12	60.63	87.43	59.16	0.693
	ESTARFM	**89.09**	**90.28**	87.62	**76.67**	**1.03**
C	FSDAF	79.02	59.46	86.56	50.27	0.687
	STARFM	79.54	54.03	83.12	47.51	0.650
	ESTARFM	**88.13**	**84.23**	**89.19**	**77.45**	**0.944**

that ESTARFM has a lower omission error and can correctly identify more snow pixels. However, the UA of the ESTARFM model in Plots A and B is 3.68%–4.36% lower than the other two methods, which is due to the model identifying more snow, thus resulting in a higher commission error. In addition, the ESTARFM model yields better Kappa compared to the other two methods, further confirming its better performance.

B. Snow Cover Distribution

The snow distribution in the study area, as derived from simulated images, is shown in Fig. 9. The study area experienced a number of snow accumulation and melting processes from January 26 to May 2, with an overall fluctuating downward trend (Fig. 9). Specifically, the snow extent gradually decreased from January 26 to February 26; remained stable from February 26 to February 29; gradually increased from March 1 to March 14; gradually decreased from March 15 to March 28; gradually increased from March 29 to April 4; and fluctuated and decreased from April 5 to May 2, with only a slight increase in some areas between April 19 and April 25. The snow extent fluctuated and decreased between April 5 and May 2, with only a slight increase in some areas between April 19 and April 25. In addition, the increase in snow extent in different time periods varied across different time periods. For example, the increase in snow extent from March 8 to March 14 occurred in the northern and southern regions of Tuole Mountain, whereas the increase from March 29 to April 4 was confined to the southern region of Tuole Mountain. Furthermore, low-elevation areas exhibited more frequent changes in snowmelt, while high-elevation areas showed more stable snow extents.

C. Evaluation of Wet Snow Identification

The study area features a complex topography with significant variations in LIA (1°–114°). The R_{VV} of wet and nonwet snow can be very similar in the extreme LIA region, which affects the identification of wet and dry snow [56]. This study analyzed the trend of wet snow R_C with LIA between 15° and 80° during the period of maximum wet snow extent (from March 8 to March 15), as backscatter values outside this range are considered unreliable [56]. As shown in Fig. 10, R_C decreases and then gradually increases as the LIA decreases, but it remains far from the −2-dB threshold. Therefore, the SAR multitemporal, multipolarization change detection algorithm can accurately identify wet snow and is more sensitive to wet snow in regions with extreme LIA compared to traditional methods [38].

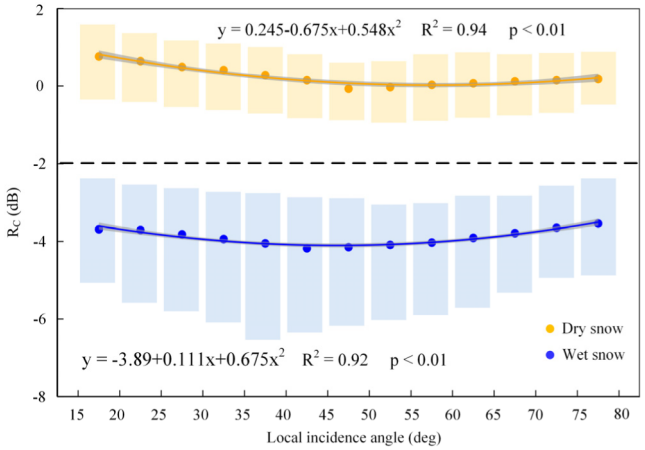

Fig. 10. Trends in R_C change with LIA from March 8 to March 15, 2020. The middle line is the median, and the upper and lower boundaries of the blue area correspond to the third and first quartiles.

D. Wet and Dry Snow Distribution

Fig. 11 illustrates the distribution of dry and wet snow in the study area from January 26 to May 2, 2020. The dry and wet snow extent within the study region exhibited significant fluctuations over time. During multiple snowmelt events, wet snow was mainly concentrated in lower altitude areas, where it melted rapidly, while dry snow was primarily distributed in higher altitude areas. As temperatures increased, snowmelt also began to occur in higher altitude regions. As shown in Fig. 12, changes in dry and wet snow extent were negatively correlated with temperature variations. The variation in wet snow areas was positively correlated with snow cover changes, indicating that snow in the lower altitude areas of the study area melted in a short period. There were no instances where wet snow areas increased due to rising temperatures, while the SCA remained unchanged, suggesting that the snow layer in these regions was relatively thin. In summary, as temperatures fluctuated and increased, the SCA in the study area gradually decreased. During the study period, the region experienced multiple cycles of snow accumulation and melting, with snow in most of the lower altitude areas being relatively shallow and exhibiting rapid changes in snow cover.

V. DISCUSSION

A. Advantages Compared to Previous Research

Compared to existing studies that utilize the ESTARFM fusion algorithm for monitoring dry and wet snow extent, the method proposed in this study offers several advancements. First, due to the correction from Landsat-8 OLI data to Landsat-7 ETM+ data (Section II-B), the proposed method is also applicable to Landsat-7 ETM+ data. This method

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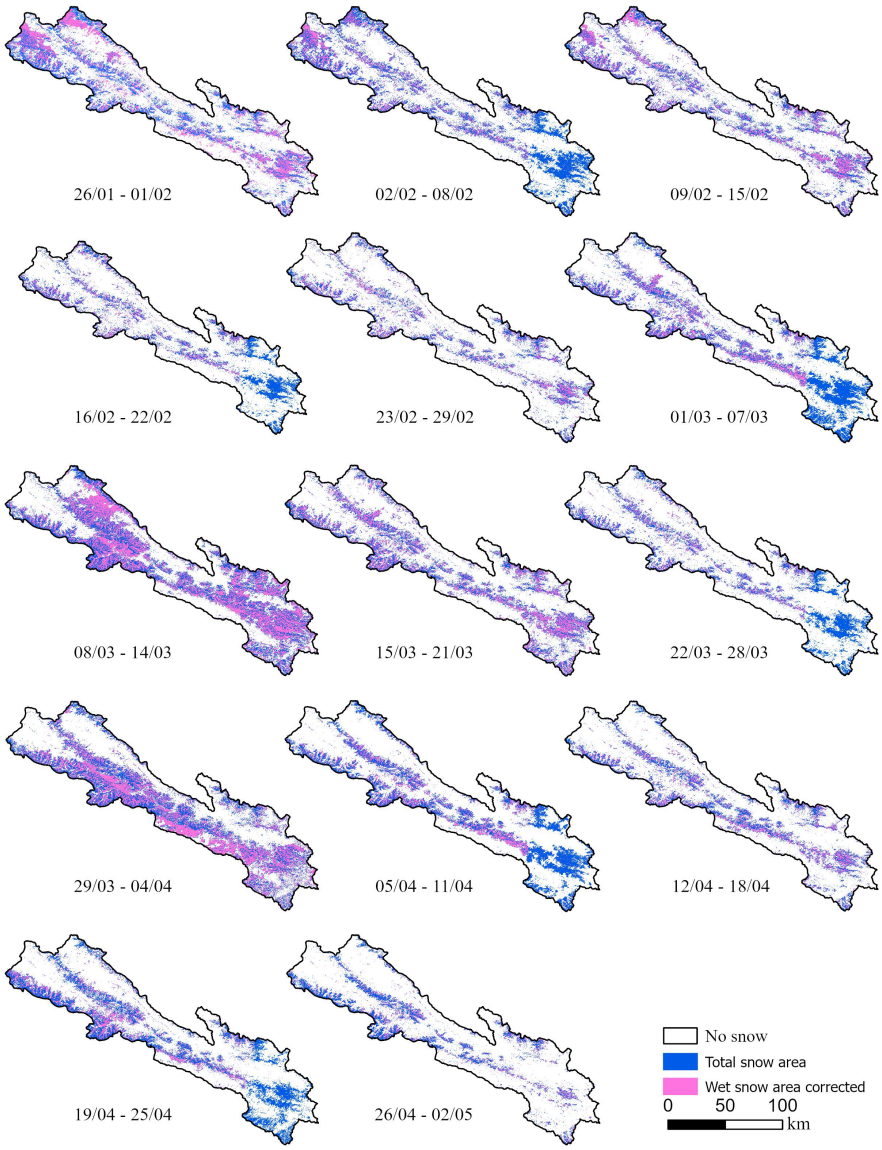

Fig. 11. Corrected dry and wet snow distribution in the study area from January 26 to May 2, 2020, based on Sentinel-1 and simulated Landsat-8 data.

can be applied to long-term studies of dry and wet snow area changes, addressing the limitations of previous research where the length of time series data available for hydrological modeling was constrained. Furthermore, with the successful launch of Landsat-9 in 2021, this method is expected to have broader applications in the future with necessary adjustments [30]. In addition, compared to previous studies that limited the monitoring of dry and wet snow extent to small river basins or valleys [53], [60], this study facilitates higher spatial and temporal resolution monitoring of dry and wet snow extent changes in large regions with complex terrain.

The proposed method can be rapidly applied to other regions for high-frequency detailed monitoring. On one hand, GEE is designed for global-scale analysis, allowing for the rapid acquisition of data such as MOD09GA, Landsat-8 OLI, and Sentinel-1 SAR. On the other hand, key steps in this study, such as correcting Landsat-8 OLI data to Landsat-7 ETM+ data and calculating LIA for the study area, can all be implemented using GEE. Moreover, the ESTARFM fusion algorithm can also be executed within GEE [80]. Therefore, if the method proposed in this study is to be applied globally, the simulated Landsat-8 OLI and Sentinel-1 SAR data can be used to generate high spatial and temporal resolution wet and dry snow extent maps for the vast majority of the world (even the polar regions) by simply substituting different DEMs (Copernicus DEM and TanDEM-X).

B. Impact of Using a Time Reference Image

Fig. 13 illustrates the influence of selecting different dry snow reference images on wet snow extent identification. Using Sentinel-1 ascending orbit data as an example, the four estimated wet snow coverage curves show a consistent overall trend, indicating that the wet snow detection results based on the SAR multitemporal, multipolarization change detection algorithm are robust, regardless of the specific dry

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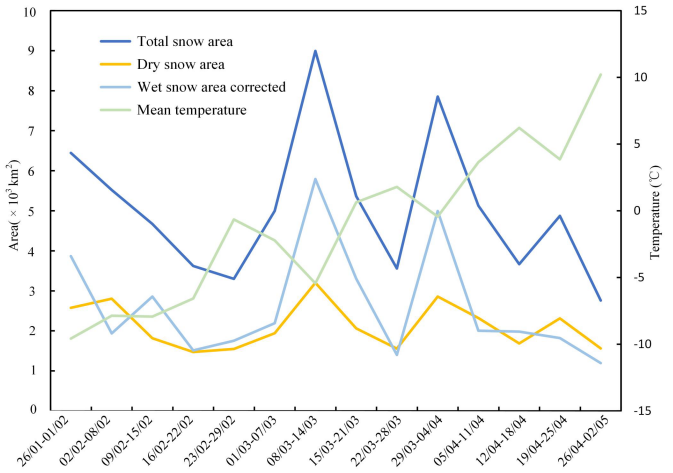

Fig. 12. Change in dry and wet snow extent per seven days from January 26 to May 2, 2020, in Qilian County.

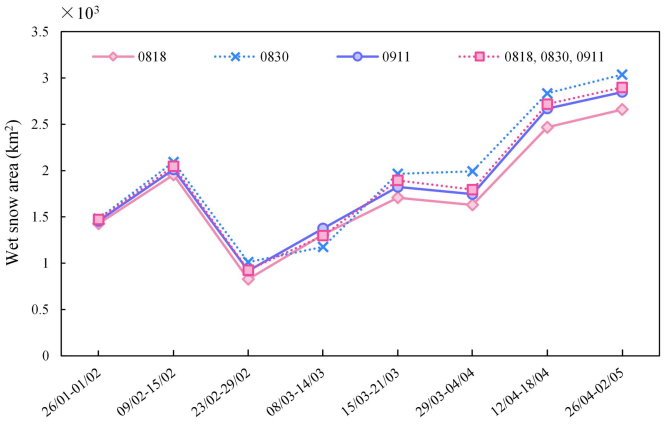

Fig. 13. Wet snow cover derived from different reference images (ascending orbit example).

snow backscatter values used as reference. Notably, when the reference backscatter is calculated as the mean of all available dry snow images from August 18 to September 18, the resulting wet snow estimation curve remains highly consistent with those derived from single-date References throughout the time series. This further demonstrates that combining multiple snow-free images improves the consistency and reliability of wet snow detection[30], [56].

Previous studies have shown that relying on a single reference image may introduce detection errors [75]. On one hand, natural temporal variations in backscatter coefficients—due to changes in soil moisture or microclimatic conditions—can lead to inconsistencies in the reference background. On the other hand, inherent speckle noise in SAR images may interfere with the accuracy of wet snow classification. Using the mean backscatter value from multiple dry snow images not only effectively suppresses random fluctuations in the time series but also reduces the influence of speckle noise, thereby improving the stability and resilience of the change detection algorithm.

In addition, seasonal vegetation dynamics significantly affect the accuracy of wet snow detection. For example, the

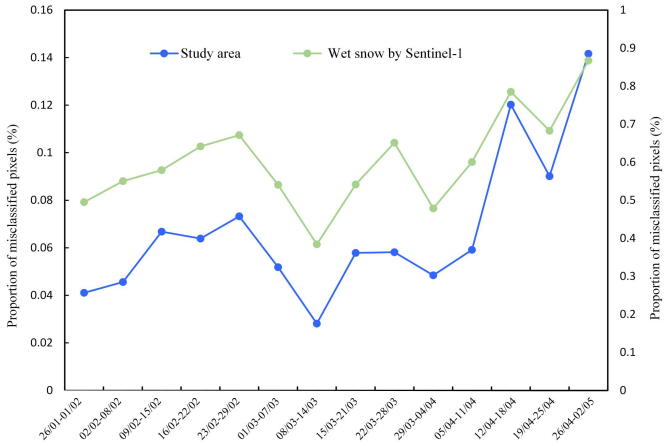

Fig. 14. Misclassified pixels as a proportion of the study area and as a proportion of the wet snow area acquired by Sentinel-1.

period from late August to early September marks the end of the growing season for herbaceous and shrub vegetation in mountainous regions. During this time, vegetation remains relatively dense and soil moisture levels are high, leading to increased surface radar backscatter. However, during the snow season (winter and spring) or snowmelt season (late spring), vegetation wilts and soils may freeze, resulting in substantial changes to radar scattering characteristics. Thus, even if the ground is snow-free during both periods, the backscatter signature can vary significantly, potentially leading to the misclassification of vegetation-induced backscatter enhancement as wet snow in change detection analyses.

Furthermore, the seasonal response of topography must be carefully considered. In mountainous regions with complex terrain—such as north-facing slopes or shaded areas where moisture tends to accumulate—stronger backscatter signals may occur in late summer and early autumn. In contrast, during winter or early spring, snow cover or frozen ground alters the surface dielectric properties, leading to significant changes in radar scattering behavior. These topography- and season-dependent variations in backscatter introduce uncertainty into multitemporal change detection and may degrade classification accuracy. Therefore, such effects should be accounted for and appropriately mitigated during algorithm development and implementation.

C. Sources of Misclassification

By calculating the wet snow pixels outside the snow cover from Sentinel-2, we can quantify the pixels misclassified as wet snow by Sentinel-1 [30]. As shown in Fig. 14, these misclassified wet snow pixels account for only a small portion of the study area (worst 0.14%, blue curve), which is due to the accurate identification of wet snow extent using the multitemporal, multipolarization SAR change detection algorithm. According to Sentinel-2 land cover data, the false alarm rate increased overall from late April to early May (blue curve), mainly in grassland areas. In addition, the aboveground biomass of natural grasslands in Qilian County begins to increase in early May [61]. Therefore, we believe that with rising temperatures, vegetation growth reduces the surface

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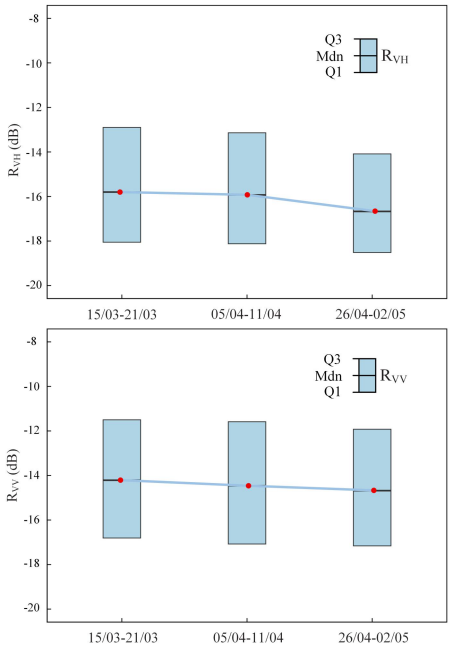

Fig. 15. Trends in misclassification of pixels from March 15 to May 2, 2020.

backscatter intensity, leading to an increase in misclassified pixels.

Further analysis revealed that starting in May, the aboveground biomass of natural grasslands in Qilian County begins to increase. The grassland types in Qilian County are mainly temperate grasslands and alpine meadows, with dominant vegetation including *Elymus*, *Stipa breviflora*, *Stipakrylovii*, and *Kobresia*, all of which are narrow-leaved plants [81]. Previous studies have demonstrated that as the leaf area index (LAI) of narrow-leaved plants increases, both co- and cross-polarized backscatter intensities tend to decrease [82]. We further analyzed the trends of R_{VH} and R_{VV} from late March to early May. The results revealed that both R_{VH} and R_{VV} exhibited a decreasing trend after late March (Fig. 15), which is consistent with previous research. We also concluded that from late April to early May, the rise in temperature led to vegetation growth, which in turn caused the simultaneous reduction in R_{VH} and R_{VV}, thereby increasing the number of misclassified pixels.

Wet snow detection in forested areas remains a significant challenge. These regions are typically concentrated in specific topographic locations, such as river valleys, north-facing slopes, and shaded mountain gullies, where limited solar radiation and lower wind speeds result in longer snow retention and different snowmelt dynamics compared to open areas. These characteristics make it difficult for change detection algorithms to accurately distinguish snow states [27], [30], [83]. Forested areas that are not masked may be systematically misclassified as dry snow or entirely omitted, especially in complex and highly shaded terrains, potentially introducing systematic errors at the subregional scale.

In this study, forest cover accounts for only 0.9% of the total study area, and thus, no masking of forested regions was applied. However, despite their limited extent, local disturbances from forested areas in topographically sensitive zones may still exert a nonnegligible influence on wet snow detection results. In future research, particularly in regions with extensive forest coverage such as northeast China, further studies on wet and dry snow discrimination are recommended. Incorporating auxiliary data such as canopy height models (e.g., LiDAR) may support model refinement and help reduce the potential impact of forests on the accuracy of wet snow identification.

D. Impact of Satellite Temporal Resolution

Although the method proposed in this study enables high-frequency, fine-scale snow cover monitoring over large mountainous areas, there remains a degree of uncertainty in distinguishing between wet and dry snow extents. One of the main challenges lies in the temporal resolution mismatch between Landsat-8 (with a 16-day revisit cycle) and MODIS (daily acquisition). Within a 30- and 45-day period, Landsat-8 typically provides only 2–3 images, while MODIS offers daily observations. This discrepancy in temporal coverage inevitably leads to biases between the multiday average snow cover products of the two sensors over the same period. However, the seven-day average imagery generated by the ESTARFM fusion model closely resembles actual observations, suggesting that the model accounts for multiday mean differences between Landsat-8 and MODIS when establishing pixel relationships, thereby producing relatively reliable results.

Sentinel-1A 12-day revisit cycle poses limitations for short-term snow condition monitoring. Within a seven-day window, typically only one SAR scene is available—sometimes none—reducing the ability to detect rapid snow cover changes. To improve observation frequency, this study incorporated descending orbit data from Sentinel-1 to fill in image gaps and enhance the temporal continuity of SAR observations. However, the descending pass occurs at approximately 23:27 local time, which is not ideal for detecting snowmelt. Snow often refreezes overnight, potentially leading to an underestimation of wet snow areas. In addition, the descent channel does not fully cover the entire study area, and the wet snow area is underestimated (Fig. 11). Future studies could address this limitation by using data from satellites with local overpasses near noon or by integrating temperature measurements from meteorological stations to correct for the underestimation of wet snow.

In addition, the lack of temporal alignment between SAR and optical remote sensing data introduces further uncertainty in wet and dry snow discrimination. Although the simulated Landsat-8 imagery generated by the ESTARFM model provides reliable seven-day average estimates of snow cover, the sparse temporal coverage of Sentinel-1 imagery may not always capture corresponding variations in wet snow accurately. This limitation is particularly pronounced when wet snow area changes significantly within the seven-day period—for instance, as seen in the increase in misclassified wet snow pixels after mid-to-late April (Fig. 13).

E. Importance of Wet Snow Correcting

Fig. 11 highlights a significant difference in wet snow extent before and after correction, primarily attributed to the

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selection of reference images. The complex terrain and elevation variations in the study area significantly influence temperature, resulting in distinct temperature distributions and pronounced vertical gradients within the region [62], [84]. Since there is no period of completely dry snow or entirely snow-free conditions, this study utilized imagery from a nearly snow-free period (from August 30 to September 6, 2019) as the reference for extracting wet snow extent. Despite this period being nearly snow-free, substantial elevation differences mean that some snow persists in high-elevation areas, while significant vegetation remains in low-elevation areas. As noted in Section V-C, the predominant vegetation in the study area's grasslands consists of narrow-leaved plants, which reduce both VV and VH polarization values. This results in the SAR detection algorithm's inability to identify wet snow in these regions [70], [72]. According to MODIS land cover data and Fig. 11, the unrecognized wet snow is predominantly located in the low-elevation grasslands. In the future we plan to combine other wet snow recognition algorithms, such as the triangulation method, to improve recognition accuracy[4].

VI. Conclusion

Monitoring the extent of dry and wet snow is crucial for understanding energy, environmental, and climate effects, as well as for hydrological modeling, water resource management, and hydropower development. This study presents a method for estimating the seven-day mean dry and wet snow extent with a 30-m resolution using the ESTARFM model, combined with MODIS, Landsat-8 OLI, and Sentinel-1 SAR data and the GEE platform. The main conclusions are as follows.

1) We proposed a method for high spatiotemporal resolution wet and dry snow identification over large and topographically complex areas, based on the ESTARFM model, multisource optical data, and active microwave radar data. The method demonstrated high accuracy.
2) We conducted an in-depth analysis of the limitations and challenges associated with the proposed approach from multiple perspectives, providing valuable insights and references for future related research.
3) The method can be implemented on the GEE platform, offering strong operability and scalability, which facilitates its potential for broader application in real-world scenarios.

This study successfully achieves continuous monitoring of snowmelt at higher spatial and temporal resolutions, effectively addressing the challenge of mapping dry and wet snow cover over large regions. Moreover, the method can be applied to any area covered by MODIS, Landsat-8 OLI, and Sentinel-1 SAR data, enabling snowmelt monitoring at a 30-m spatial resolution and seven-day temporal resolution. The resulting dataset contributes to a better understanding of snowmelt and hydrological processes and provides scientific support for research on regional energy balance, climate, and water cycles.

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